Ben Robison

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Education

University of Washington – BS in Electrical and Computer Engineering June 2028 (Expected)

- GPA: **3.8**; Dean's List (six quarters); Tau Beta Pi Engineering Honor Society

Skills

RTL/HDL: SystemVerilog, Verilog, VHDL; RTL design and verification, FPGA synthesis, static timing analysis
Programming C, Python, Java, SQL, Bash, Git; Linux, embedded Linux
Hardware/EDA: Quartus Prime, TimeQuest, Efinity, ModelSim, Cadence Virtuoso, HSPICE, KiCad; oscilloscopes

Projects

FPGA GEMM Accelerator ae0lis.github.io/projects/gemm_accelerator

- Designed an 8×8 signed-INT8 GEMM accelerator in SystemVerilog using 64 parallel MAC units with INT32 accumulation; synthesized to 64 Cyclone V DSP blocks.
- Integrated the accelerator with the ARM HPS over Avalon-MM and wrote an embedded Linux C driver to load operands, control execution, and compare all 64 outputs against a software reference on a DE1-SoC.
- Computed one 8×8-by-8×8 matrix product in nine core cycles (180 ns at 50 MHz, excluding host transfers) with a 77.48 MHz post-fit worst-corner Fmax.
- Verified the RTL with self-checking testbenches and 1,000 randomized control-interface operations.

FPGA Rhythm Game ae0lis.github.io/projects/dwyt

- Built a four-lane rhythm game in SystemVerilog on a Cyclone V FPGA, using per-row FSMs and 10-bit LFSRs to generate and score notes across four difficulty levels on a 16×16 LED matrix.
- Implemented a single-clock enable architecture, two-flip-flop input synchronizers, and event latching; reduced the initial logic footprint by approximately 35% to 330 ALUTs and 194 registers.

Experience

Undergraduate Researcher, ACME Lab @ UW (Prof. Scott Hauck) – Seattle, WA Sep 2026 – Present

- Invited by Prof. Hauck to join a six-person team of graduate students and industry engineers developing FPGA models that locate Gaussians in high-speed camera footage.
- Learned VHDL to adapt a detection design from 5 to 3 Gaussians; cut a 9-state FSM to 8, saving a flip-flop.

FPGA/Embedded Systems Engineer, Husky Robotics @ UW – Seattle, WA Feb 2026 – Present

- Building an APB3 slave exposing PWM period and duty-cycle registers to an Efinix Ti375 RISC-V SoC.
- Automated the rover's soil-sampling sequence in C on a Raspberry Pi, coordinating auger, lift, and collection-box actuation with time-of-flight fill detection and Hall-effect homing and dig-stop limits.
- Refactored the team's shared C codebase with conditional compilation and Makefile targets for reuse.

Summer Associate, rehired as Program Analyst, CBRE (Google Account) – San Jose, CA Jun 2026 – Sep 2026

- Designed a SQL KPI dashboard; presented its design to a senior Google stakeholder and won approval.
- Audited an engineering report guiding facilities strategy, avoiding ∼$10K in unnecessary work.

Technical Writer, SEAL Lab @ UW – NSF, SBIR, and NIH proposal writing Dec 2025 – Apr 2026

Leadership & Activities

Vice President & Assistant Instructor, UW Historical Fencing – Seattle, WA Sep 2025 – Present

- Coaching beginners one-on-one in longsword and saber; helped grow first-meeting turnout from ∼12 to 30+.
- Helped revive the club: set up its bank account, recruited officers, ran events, and managed sponsor relations.